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Where Minds Pollinate

Class 10

Biology

Chapter 3

Short Questions Notes

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BIOLOGY

CLASS: 10TH

CHAPTER 3: RESPIRATORY SYSTEM

SHORT ANSWER QUESTIONS :

1. How does the exchange of gases take place in alveoli?

The **alveoli** are tiny air sacs in the lungs surrounded by capillaries. When we breathe in, oxygen in the alveoli moves into the blood through the thin walls by diffusion because the oxygen concentration is higher in the **alveoli** than in the blood. At the same time, **carbon dioxide**, which is higher in concentration in the blood, diffuses out into the alveoli to be exhaled. This gas exchange is very efficient because the alveoli have a large surface area, thin walls, and are moist.

2. What is the effect of exercise on the rate of breathing?

When we exercise, our muscles need more energy. To produce this energy, the body needs more **oxygen** and must get rid of extra **carbon dioxide** produced. This makes the brain send signals to increase the rate and depth of breathing. As a result, we breathe faster and deeper during exercise to meet the **oxygen** demand and remove **carbon dioxide** quickly.

3 Compare inspired air and expired air.

Inspired air is the air we take in from the atmosphere. It contains about 21% oxygen, 0.04% carbon dioxide, and very little water vapour. Expired air, which we breathe out, has about 16% oxygen because some is used by the body, about 4% carbon dioxide because the body produces it, and a lot of water vapour. This shows how the lungs exchange gases during breathing.

4. How do viral infections like COVID-19 affect the bronchi?

Viral infections such as COVID-19 can cause inflammation of the bronchi, which are the large airways that carry air to the lungs. The inner lining of the bronchi becomes swollen and may produce extra mucus, which blocks the airways. This leads to coughing, shortness of breath, and chest tightness. In severe cases, the infection can damage lung tissues and make breathing very difficult.

5. What is the importance of breathing in fresh air?

Breathing in fresh air is important because it contains a good amount of oxygen, which our body needs to produce energy. Fresh air is usually free from harmful gases, dust, and smoke, so it keeps our lungs healthy and reduces the risk of respiratory diseases. It also helps in better brain function and keeps us active and refreshed.

6. What is the importance of keeping the nasal and oral cavity clear?

Keeping the nasal and oral cavities clear is important because they are the main pathways for air to enter the body. If they are blocked by mucus, dust, or infections, it becomes difficult to breathe properly. A clean nasal cavity filters dust and germs, warms and moistens the air, and protects the lungs from harmful substances.

7. Differentiate between:

a. Respiration and Breathing

Breathing is the physical process of taking in oxygen and releasing carbon dioxide through the lungs. Respiration is a chemical process where oxygen is used to break down food (glucose) inside cells to produce energy.

b. Respiration and Photosynthesis

Respiration happens in all living things to release energy by using oxygen and glucose. **Photosynthesis** only occurs in plants and some bacteria to make food (**glucose**) using carbon dioxide, water, and sunlight.

c. Inhalation and Exhalation

Inhalation is the process of taking air into the lungs. The **diaphragm** contracts and moves down. **Exhalation** is pushing air out of the lungs when the diaphragm relaxes and moves up.

d. Smoker and Passive Smoker

A **smoker** actively inhales tobacco smoke by smoking cigarettes. A **passive smoker** is a non-smoker who breathes in smoke from people smoking nearby, which can also harm their lungs.

e. Bronchitis and Pneumonia

Bronchitis is inflammation of the bronchi, causing cough and mucus. **Pneumonia** is an infection where air sacs (alveoli) fill with fluid or pus, making breathing very hard.

f. Aerobic and Anaerobic respiration

Aerobic respiration uses oxygen to produce energy from food. Anaerobic respiration does not use oxygen and produces less energy, with lactic acid or alcohol as by-products.

g. Pharynx and Larynx

The pharynx is a muscular tube that serves as a passage for food and air. The larynx, also known as the **voice box**, is a part of the airway that helps produce sound.

h. Glottis and Epiglottis

The **glottis** is the opening between the vocal cords in the larynx. The **epiglottis** is a flap of tissue that closes the glottis during swallowing to prevent food from entering the airway.

i. Bronchi and Bronchioles

Bronchi are large air tubes that branch from the trachea into each lung. Bronchioles are smaller branches of the bronchi that end at the alveoli, where gas exchange occurs.

j. Air sacs and Alveoli

Air sacs are clusters of alveoli. Alveoli are tiny, balloon-like structures where the actual exchange of oxygen and carbon dioxide takes place.

k. Oxyhaemoglobin and Carboxyhaemoglobin

Oxyhaemoglobin is formed when oxygen binds with haemoglobin in red blood cells. Carboxyhaemoglobin is formed when carbon monoxide binds with haemoglobin, which is harmful because it prevents oxygen transport.

8. Breathing through the nose is healthier than breathing by mouth. Why?

Breathing through the nose is healthier because the nose filters dust, germs, and other harmful particles using tiny hairs and mucus. It also warms and moistens the air before it reaches the lungs. Mouth breathing does not provide these protections and can lead to a dry throat and infections.

9. What are the conditions necessary for efficient gaseous exchange between an organism and the environment?

For efficient gaseous exchange:

- There must be a large surface area (like alveoli) for gases to pass through.
- The surface should be thin so gases can diffuse easily.
- The surface must be moist for gases to dissolve.
- There should be a good blood supply to carry gases.
- There should be a difference in concentration of gases (oxygen is high outside, carbon dioxide is high inside) to allow diffusion.

10. Which diagram, A or B, shows the process of exhalation? Support your answer with a reason.

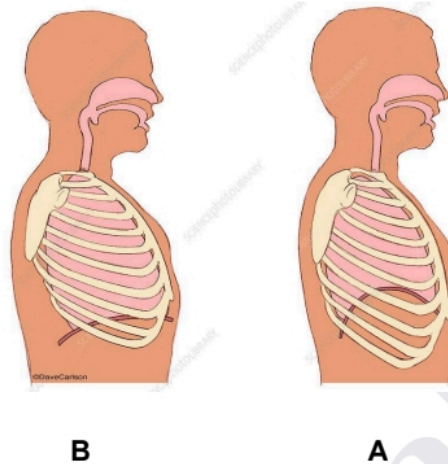
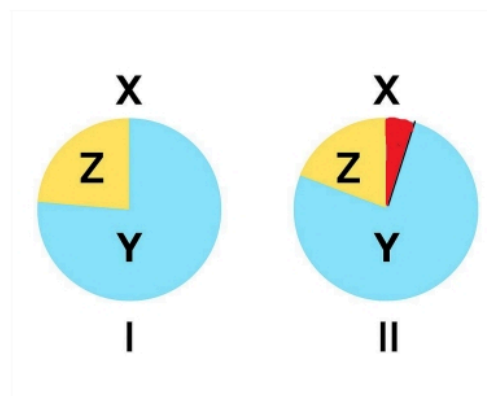


Diagram B shows exhalation. This is because the diaphragm is shown in a relaxed, dome-shaped position, and the rib cage looks smaller, meaning the chest cavity volume has decreased, pushing air out of the lungs.

11. Describe the effects of cigarette smoking on the inner walls of trachea of human beings.

Cigarette smoke damages the inner lining of the **trachea**. It destroys cilia (tiny hair-like structures) that help clean the airways by moving mucus and trapped particles out. This leads to a buildup of harmful substances, more infections, and diseases like bronchitis and lung cancer. The smoke also irritates the lining, causing inflammation and excess mucus.

12. The diagram shows the percentage of gases in inspired and expired air.



a. Identify I, II, X, Y and Z.

- I = Inspired air
- II = Expired air
- X = Carbon dioxide
- Y = Nitrogen
- Z = Oxygen

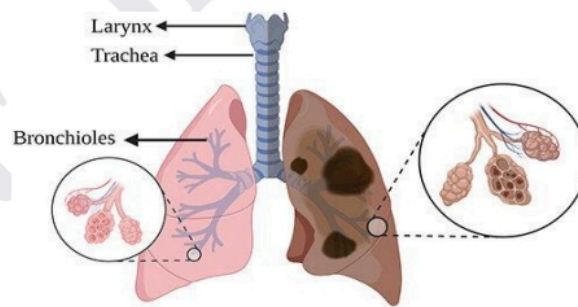
b. Why is gas X almost negligible in Figure I?

Gas X (carbon dioxide) is almost negligible in inspired air because the atmosphere has a very small amount of carbon dioxide (**about 0.04%**).

c. How is the percentage of X increased in Figure II?

The percentage of gas X (**carbon dioxide**) increases in expired air because carbon dioxide is produced as a waste product during respiration in the body and is removed through exhalation.

13. The figure shows a healthy lung and a diseased lung



a. Identify the disease and the affected part of the lung.

The disease shown is **Emphysema**, which affects the **alveoli** — the tiny air sacs in the lungs where gas exchange occurs.

b. How does part B show structural variation from part A?

In part A (healthy alveoli), the alveoli are small, round, and clustered closely together, which

provides a large surface area for gas exchange.

In part B (diseased alveoli), the walls of the alveoli are broken down, creating fewer and larger air spaces. This reduces the surface area, making gas exchange less efficient.

c. Write any two symptoms of the disease.

1. **Shortness of breath**, especially during physical activity.
2. **Chronic cough** with or without mucus.

14. The diagram shows the structural and functional units of the lungs

a. Identify structure Z.

Structure Z is a **capillary**, a tiny blood vessel that surrounds the alveolus to allow exchange of gases between air and blood.

b. Name the gases X and Y.

- Gas X is **oxygen (O_2)**, which enters the blood from the alveoli.
- Gas Y is **carbon dioxide (CO_2)**, which moves from the blood into the alveoli to be exhaled.

c. What is the name of the blood which is leaving at point E?

The blood leaving at point E is called **oxygenated blood** because it has received oxygen from the alveoli.

15. How do the nose and the nasal cavity function in filtering incoming air?

The nose and nasal cavity contain **tiny hairs and mucus**. These trap **dust, bacteria, and other particles**, preventing them from entering the lungs. The nasal passages also **warm and moisten** the air, making it easier for the lungs to absorb the gases.

16. What is the role of the pharynx in human respiration?

The **pharynx** is a part of the throat that acts as a **passageway for both air and food**. During respiration, it helps **channel air from the nose or mouth to the larynx (voice box)** and then into the trachea (windpipe). It also plays a role in protecting the airway during swallowing by preventing food from entering the respiratory tract.

